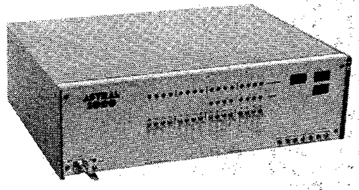
NEW MICROCOMPUTER

At last, the much talked about **Astral 2000** microcomputer is ready to go. If you were at the last meeting of the **Homebrew Computer Club**, you were among the first to see an actual demonstration of the **Astral** given by **Marty Spergel** from **M&R Enterprises**, the manufacturer of the **Astral**.

The **Astral** is a significant feather in the cap of the **Homebrew Computer Club** for it has been designed and manufactured entirely by club members. The chief engineer for the **Astral** project is **Carl Kelb** who operates his own consulting firm, **R C Engineering Co.**

The **Astral 2000** is an extremely powerful micro, so powerful that **Carl Helmers** of **Byte** magazine has described it more as a *mini* than as a micro in terms of capabilities and exapandability.



Astral 2000 Microcomputer

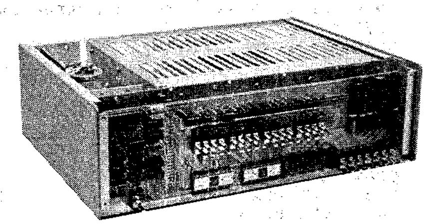
The system is housed in a well-built, professional quality cabinet and incorporates a modular power supply from PowerTec which is quite adequate for the job. An interesting item to note about the assembly of the **Astral** is that there is not a single wire in the entire machine. Correction: there is one actually—the line cord. All the other components plug directly into an expanable system bus, including the front panel assembly. Although the **Astral** is advertised as a "kit", in reality it is 90% assembled upon delivery. All boards are fully stuffed, tested and burned in for a minimum of 24 hours. The only assembly procedures required are simply putting the cabinet together, placing the power supply inside and bolting it down, attaching the line cord and inserting the various circuit boards.

The front panel assembly contains the usual sets of LEDs and switches, however there are a few interesting features which should be pointed out. A real time clock driven by an integrated clock chip reads out time in hours, minutes and seconds. The same displays used for the clock can also be used to read out programs in hex, an invaluable debugging aid. In addition, the front panel is attached to the back plane by a special connector arrangement designed in such a way that the two assemblies simply are snapped together. Remember putting together that incredible wiring harness for your **Altair**? Well, rest assured that will never happen with the **Astral**.

Three circuit boards are currently available for the system. The processor board contains a 6800 MPU and a great number of other devices as well. The processor operates both in serial and parallel. The serial I/O port outputs both RS-232 and 20mA current loop for teletypes. The serial I/O port is fully protected by opto-isolators. The processor board is shipped with Motorola's MIKBUG monitor and 384 bytes of 6810-1 RAM installed.

Two types of memory boards are in production for the system. The first is the 8K RAM board which utilizes low power static RAMs with a 500ns cycle time. The entire 8K RAM board draws a mere 1.5A from a single +5V supply. With the MIKBUG installed, only five RAM boards can be adressed by the processor, however without MIKBUG, the 6800 can talk to 65K of memory (8 RAM boards). The location of each RAM board in the memory map is selected by a set of jumpers on the RAM board itself.

And for you people out there who are drowning in paper tape, an 8K EPROM board is available for either the Motorola or the AMI 5204 erasable PROMs. This board is fully stuffed with all the miscellaneous control and decode logic, however it does not contain the EPROMs although sockets for the EPROMs are furnished.



Astral 2000 With Front Panel Cover Removed

A video display module has already been designed by our venerable club leader, **Lee Felsentein** who is also responsible for the design of both the **VDM-1 (Processor Technology)** and the **Pennywhistle 103** acoustic coupler **(M&R Enterprises again)**. This new display module (the **VID-80**) has been designed specifically for the **Astral** and—of course—plugs directly into the system bus with no further ado. The **VID-80** offers selectable line length with adjustments for 64, 72 and 80 characters per line. The **VID-80** will display up to 24 lines of upper and lower case characters.

Other future additions to the system will include a number of inexpensive peripherals. Although Marty is still negotiating with various manufacturers, he does expect to be able to offer reasonably priced digital tape decks, tape readers and floppy disks with controllers implemented on **Astral** bus-compatible cards.

Initially, software will include a special version of BASIC designed specifically for operating in this system. BASIC will be available both in tape and in PROM.